Why a Scalar Potential Fails for Oscillating Fields

And why we pivoted to a vector potential

MTP — Negative rotational viscosity in ferrofluids via phase-field MRT-LBM

Read this first

We want a magnetic field that points *down the channel* (along x) and that we can switch on and off in time (AC). The natural first choice is a *scalar potential* ψ, with $\mathbf{H} = -\nabla\psi$. It is one number per grid point, cheap, and solves a plain Poisson equation. It *fails* — not because of a bug, but because of the *shape of the channel*. The fix, a *vector potential* A_z, is *also just one number per grid point* and solves the *same* Poisson equation, but it stores the field in a way the geometry cannot erase. This PDF explains why, from the ground up.

1 The setup, stripped to essentials

The magnetic problem in the fluid is quasi-static: the field oscillates slowly compared with how fast light crosses the domain, so displacement currents vanish and at *every instant*

$$\nabla \times \mathbf{H} = 0, \qquad \nabla \cdot \mathbf{B} = 0, \qquad \mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}). \tag{1}$$

Time does not appear in these equations. It enters *only* through the boundary data. So an oscillating field is just a static-field problem re-solved every timestep with the boundary values scaled by some $s(t)$ (e.g. $\cos\omega t$). This is the crucial fact: **the difficulty is not "time" at all — it is whether the geometry lets a field with a nonzero average along x exist on a channel that is periodic in x.**

Why the channel is periodic in x

The channel is a slab: walls at $y = 0$ and $y = H$, and we tile it in x so a particle leaving the right edge re-enters the left. That periodicity is what makes it a "channel" rather than a finite box, and it is exactly the property that breaks the scalar potential.

2 The scalar potential, and the trap it walks into

Curl-free means $\mathbf{H}$ is a gradient. Write $\mathbf{H} = \mathbf{H}_{\text{ext}} - \nabla\psi$ or, in the pure form, $\mathbf{H} = -\nabla\psi$. Feeding this into $\nabla \cdot \mathbf{B} = 0$ gives the Poisson problem your ħ-lattice originally solved:

$$\nabla^2\psi = \nabla \cdot \mathbf{M}. \tag{2}$$

The landscape picture — this is the whole argument

Think of ψ as the **height of a hilly landscape**, and $\mathbf{H} = -\nabla\psi$ as the **downhill slope** a ball would roll. A field pointing steadily along $+x$ means the ground slopes downhill the *entire way* across the channel in x.

Now walk one full lap around the channel in x and come back to where you started. Because the channel is periodic, *you are at the same point*, so you must be at the *same height*: $\psi(x + L, y) = \psi(x, y)$. But if the ground sloped downhill the whole way, you would have descended — you would end *lower* than you began. Contradiction. A closed loop cannot go downhill all the way round.

So on a periodic channel, ψ *cannot* carry a net (mean) field along x. The average $\partial_x\psi$ over a period is forced to zero. **The applied down-channel field silently disappears.**

Mathematically: any periodic function has zero mean slope,

$$\frac{1}{L}\int_0^L \partial_x\psi\, dx = \frac{\psi(L,y)-\psi(0,y)}{L} = 0 \quad\Longrightarrow\quad \langle H_x\rangle_x = -\langle\partial_x\psi\rangle_x = 0. \tag{3}$$

A direct lattice test (`vecpot.py`) solved Eq. (2) on the periodic channel for a magnetization $\mathbf{M}$ carrying an intended mean field H_{0x}. The recovered mean field came back as $\langle H_x\rangle \to 0$: **the applied field was lost**, exactly as the landscape argument predicts. The same script with the vector potential (next section) recovered H_{0x} to machine precision.

Why "it's a static-field limitation" was the wrong diagnosis

It is tempting to say "scalar potentials only work for static fields." That is *false*. The scalar potential handles time fine (time is just $s(t)$ on the boundary). What it cannot handle is a **nonzero mean field along the periodic direction** — static or oscillating. The AC case merely made the failure obvious, because the drive we needed was a down-channel field. The real culprit is topology (the loop), not time.

3 The vector potential: same cost, but topology-proof

In 2D, define the field through a single scalar $A_z(x,y)$:

$$\mathbf{B} = \nabla\times(A_z\hat{z}) = \big(\partial_y A_z,\ -\partial_x A_z\big). \tag{4}$$

Two things happen for free. First, $\nabla\cdot\mathbf{B} = \partial_x\partial_y A_z - \partial_y\partial_x A_z \equiv 0$: the divergence constraint is satisfied *identically*, for *any* A_z. Second, imposing $\nabla\times\mathbf{H} = 0$ gives a Poisson equation of the *same form* as before, only the source changes from a divergence to a curl:

$$\boxed{\nabla^2 A_z = -\mu_0\,(\nabla\times\mathbf{M})_z}, \qquad \mathbf{H} = \frac{1}{\mu_0}\big(\partial_y A_z,\ -\partial_x A_z\big) - \mathbf{M}. \tag{5}$$

Why the loop no longer bites

With the scalar potential, "field along x" meant "ψ keeps dropping in x" — impossible on a loop. With the vector potential, a *uniform field along* x, $H_x = \text{const}$, corresponds to A_z that *rises linearly in* y: $H_x = \partial_y A_z/\mu_0$. The variation is in y, the *non-periodic* direction (it has walls, not a wrap). Nothing has to slope-and-return in x. So A_z can hold a steady down-channel field without any contradiction. The field is stored *across* the channel, where there is no loop to close.

The practical payoff is that the driving field enters through the **wall boundary condition** on A_z and is carried, by the solver, into the interior:

$$\left.\frac{\partial A_z}{\partial n}\right|_{\text{wall}} = \mu_0\,(\mathbf{H}_{\text{app}} + \mathbf{M}_w)\cdot\hat{\mathbf{t}} \qquad (\text{Neumann};\ \hat{\mathbf{t}} = \hat{\mathbf{n}}\times\hat{z}), \tag{6}$$

so for a down-channel applied field H_x the walls carry $\partial A_z/\partial n = \pm\mu_0(H_x + M_{x,w})$. This is a physical statement (tangential $\mathbf{H}$ continuous across the wall), not an analytic patch grafted onto the interior.

It is fair to worry: "does A_z actually *propagate* the oscillating field through the whole fluid, or does it also lose it somewhere?" It *does* propagate it. The only limit is a **causality floor**: the lattice moves information at most one site per step, so filling the domain takes at least

$$\tau_{\text{relax}}^{\min} = \frac{1}{c_s k_{\min}} = \frac{n_Y}{\pi c_s} \text{ steps.}$$

This is *not* a formulation error and *no* choice of ε, MRT rates, or boundary treatment beats it — it is set by n_Y and c_s alone. The drive must simply be slow relative to it, $\omega\,\tau_{\text{relax}} \ll 1$, which is why we **sub-cycle** the `h`-lattice ($N_{\text{sub}} = 64$) between physical steps. Rejecting the "analytically split off the applied field" shortcut was correct: that shortcut hides the failure instead of fixing it.

4 Why not store B directly? (The Dellar route)

One could store the field itself as the moment of a *vector* distribution, $\mathbf{B} = \sum_k \mathbf{g}_k$ (Dellar, *J. Comput. Phys.* **179**, 95, 2002). Because $\mathbf{B}$ is stored with no potential behind it, there is no single-valuedness constraint and it *can* carry a mean field. But it does not fit our problem:

Constraint vs. evolution

Dellar's scheme solves the **induction equation** — a genuine time-evolution of $\mathbf{B}$ — and gets $\nabla \cdot \mathbf{B} = 0$ for free *because that equation is already in curl form.* Ours is a **constraint**, $\nabla \times \mathbf{H} = 0$ and $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$, with *no* time derivative. We would have to bolt on an ad-hoc relaxation toward $\nabla \times \mathbf{H} = 0$, police it every step, run two lattices, and reuse none of our derivation. The vector potential gives $\nabla \cdot \mathbf{B} = 0$ *and* $\nabla \times \mathbf{H} = 0$ from one scalar Poisson solve.

5 The bottom line

- The scalar potential ψ loses any *mean field along the periodic (x) direction* — confirmed numerically. This is a topological fact about closed loops, not a limitation to static fields.
- The vector potential A_z is *one scalar*, solves the *same* Poisson equation (source $\nabla \times \mathbf{M}$ instead of $\nabla \cdot \mathbf{M}$; recovery $\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$ instead of $-\nabla\psi$), and stores a down-channel field as variation in the *non-periodic* y-direction, where no loop closes.
- A_z carries the AC drive through the whole fluid, limited only by the causality floor $\tau_{\text{relax}}^{\min} = n_Y/(\pi c_s)$, which sub-cycling respects.
- Storing $\mathbf{B}$ directly (Dellar) works for *induction* problems but not for our *magnetostatic constraint.*